

Generative 3D Art & Object Studio

Problem:

Creating custom 3D objects and complex 3D compositions can be difficult for beginners because traditional 3D modeling software often requires learning many tools and techniques. Even when the user only wants to create a simple design, they may need to manually create objects, position them, rotate them, resize them, duplicate them, and arrange them properly in a 3D environment. This becomes especially time-consuming when the design contains hundreds or thousands of repeated objects.

For example, imagine a user wants to create 200 spheres arranged in a spiral, with each sphere having a different size and rotation. In traditional software, setting up every object individually would be repetitive and inefficient. The user needs a simpler way to describe **what they want the computer to generate** rather than manually creating every element.

Solution:

Generative 3D Art & Object Studio is a browser-based 3D creation tool that allows users to create objects and complex 3D artwork using **basic geometric shapes, mathematical rules, and parameters**. Instead of uploading an existing 3D model, the user starts by creating the object directly inside the application.

Users can choose primitives such as **cubes, spheres, cylinders, cones, toruses, planes, and custom shapes**. They can then control properties such as size, position, rotation, scale, segments, spacing, and repetition.

The real power of the system comes from its **mathematical generation engine**. Users can select patterns such as **spiral, grid, circle, wave, radial, sphere, symmetry, or random distribution**. JavaScript calculates the position, rotation, scale, and relationship of each generated object automatically.

For example, instead of manually placing 100 spheres, the user can specify:

100 objects → Spiral → Radius 20 → Rotation 30° → Scale 0.5–2

The system calculates the required coordinates and generates the entire composition automatically.

The application can be built using **HTML, CSS, JavaScript, [Three.js](#), Python, and WebGL**. Finally, users can modify their creation, preview it in 3D, and optionally **export the generated artwork as a GLB file** for use in other 3D applications.